Øving 4

ØVING 4

1) a) $[H^+] = [OH^-]$

b) $[H^+] = [OH^-] + [Cl^-]$ ← sterk syre ⇒ $[OH^-] \simeq 0$

$[H^+] = [Cl^-]$

c) $[H^+] = [HCOO^-] + [OH^-]$

d)

e) $H_2SO_4 \rightarrow HSO_4^- + H^+$

$HSO_4^- \rightleftharpoons SO_4^{2-} + H^+$ $[OH^-] \approx [SO_4^{2-}]$

$[H^+] = [HSO^-] + [SO_4^{2-}] + [OH^-] \Rightarrow [HSO^-] + 2[SO_4^{2-}]$

2) a) $NaCl \xrightarrow{H_2O} Na^+ + Cl^-$

EN: $[Na^+] + [H^+] = [Cl^-] + [OH^-]$

MB: $[NaCl]_0 = [Na^+] = [Cl^-]$ ← pH blir ikke nøytralt

b) $HCl \xrightarrow{H_2O} H^+ + Cl^-$

EN: $[H^+] = [OH^-] + [Cl^-]$

MB: $[HCl]_0 = [H^+]$ ← HCl er en sterk syre

c) $HCOOH \rightarrow HCOO^- + H^+$

EN: $[H^+] = [HCOO^-] + [OH^-]$

MB: $[HCOOH]_0 = [HCOOH] + [HCOO^-]$

d)

OVING 4

2) a) $H_2SO_4 \rightarrow HSO_4^- + H^+$

$HSO_4^- \rightarrow SO_4^{2-} + H^+$

veldig lite

EN: $[H^+] = [HSO_4^-] + 2[SO_4^{2-}] + [OH^-] \simeq [HSO_4^-] + 2[SO_4^{2-}]$

MB: $[H_2SO_4]_0 = [H_2SO_4] + [HSO_4^-] + [SO_4^{2-}]$

3) $[Na^+] = 0{,}001\ mol\ dm^{-3}$ $[Cl^-] = 0{,}015\ mol\ dm^{-3}$ $[ClO_4^-] = 0{,}01\ mol\ dm^{-3}$

$[K^+] = [Cl^-] - [Na^+] + [ClO_4^-] = 0{,}024\ mol/dm^3$

4) Massevirkningsloven $k_a = \frac{[H^+][F^-]}{[HF]}$ $[HF]_0 = 0{,}25\ mol/dm^3$

$pk_a = 3{,}17 \rightarrow k_a = 6{,}76 \cdot 10^{-4}$

$k_w = 10^{-14}$

$[H_3O^+] = [H^+] = ?$

MB: $[HF]_0 = [F^-] + [HF]$

EN: $[H^+] = [F^-] + [OH^-] \simeq [F^-]$

$[OH^-] << [H^+]$

$[F^-] = [H^+]$ & $[HF] = [HF]_0 - [F^-] = [HF]_0 - [H^+]$

$k_a = \frac{[H^+]^2}{[HF]_0 - [H^+]} \quad | \cdot ([HF]_0 - [H^+])$

$[H^+]^2 = k_a[HF]_0 - k_a[H^+]$ ← $[H^+] = x$

$x^2 + k_a x - k_a[HF]_0 = 0$

$x = \frac{-k_a \pm \sqrt{k_a^2 - 4(-k_a[HF]_0)}}{2}$ $x_1 = 0{,}013$

$x_2 = -0{,}013$ ← gir

$pH = -\log 0{,}013 = \underline{\underline{1{,}89}}$

$6{,}76 \cdot 10^{-4}$

ÖVNING 4

5) a) $CH_3CH(OH)COOH + H_2O \leftrightharpoons CH_3CH(OH)COO^- + H_3O^+$

$HA + H_2O \leftrightharpoons A^- + H_3O^+$

$[HA]_0 = 0{,}1\ mol/dm^3 \quad pK_a = 3{,}86 \quad K_a = 1{,}380 \cdot 10^{-4}$

$K_a = \frac{[A^-][H_3O^+]}{[HA]} \qquad K_w = [H_3O^+][OH^-] = 10^{-14}$

$\hookrightarrow [OH^-] = K_w / [H_3O^+]$

EN: $[H_3O^+] = [A^-] + [OH^-] \rightarrow [A^-] = [H_3O^+] - [OH^-]$

MB: $[HA]_0 = [HA] + [A^-] \rightarrow [HA] = [HA]_0 - [A^-]$

$K_a = \frac{[A^-][H_3O^+]}{[HA]_0 - [A^-]} = \frac{[H_3O^+]([H_3O^+] - [OH^-])}{[HA]_0 - ([H_3O^+] - [OH^-])}$

$= \frac{[H_3O^+]([H_3O^+] - (K_w / [H_3O^+]))}{[HA]_0 - ([H_3O^+] - K_w / [H_3O^+])}$ } anta att $\frac{K_w}{[H_3O^+]}$ är försvinnande liten

~~$K_a = \frac{([H_3O^+])^2}{[HA]_0 - [H_3O^+]}$ Svårt syre, se antar att $[H_3O^+]$ är försvinnande liten~~

$K_a = \frac{[H_3O^+]^2}{[HA]_0 - [H_3O^+]} \rightarrow [H_3O^+]^2 = K_a[HA]_0 - K_a[H_3O^+]$

$[H_3O^+]^2 + K_a[H_3O^+] - K_a[HA]_0 = 0$

$[H_3O^+] = \frac{-K_a \pm \sqrt{K_a^2 - 4(-K_a[HA]_0)}}{2} \qquad x_1 = 3{,}646 \cdot 10^{-3}$

$x_2 = -3{,}784 \cdot 10^{-3}$ ← gränsfall

$pH = -\log(3{,}646 \cdot 10^{-3}) = \underline{\underline{2{,}43}}$

undersöker tillräcklighet

$[OH^-] = \frac{K_w}{[H_3O^+]} = \frac{10^{-14}}{0{,}0365} \approx 10^{-13}$ ← tillräcklig ok

ØVING 4

5) b) $pH = pka \pm 0,5$ $\quad [HA] = 0,1$ mol dm^{-3}

$pH = pka \Rightarrow -\log[H^+] = -\log ka \Rightarrow [H^+] = ka$

$\hookrightarrow \frac{ka}{[H^+]} = 1 = \frac{[A^-]}{[HA]}$

$ka = \frac{[A^-][H^+]}{[HA]}$ $\quad$ MB: $[HA]_0 = [HA] + [A^-] = 0,1$

$[HA] = [A^-] = 0,05$

$pH = pka - 0,5$

$-\log[H^+] = -\log ka - 0,5$

$\log[H^+] = \log ka + 0,5 \Rightarrow \log ka - \log[H^+] = -0,5$

$\log \frac{ka}{[H^+]} = -0,5 \quad | \wedge 10 \rightarrow \frac{ka}{[H^+]} = 10^{-0,5} = \frac{[A^-]}{[HA]} = 0,32$

$[HA]_0 = [HA] + [A^-] = [HA] + 0,32[HA] = 1,32[HA]$

$[HA]_0 = 0,1$

$\underline{[HA] = 0,076 \text{ mol/l}}$

$pH = pka + 0,5$

$-\log[H^+] = -\log ka + 0,5$ ← bytter fortegn

$\log[H^+] = \log ka - 0,5 \Rightarrow \log ka - \log[H^+] = 0,5$

$\log \frac{ka}{[}$